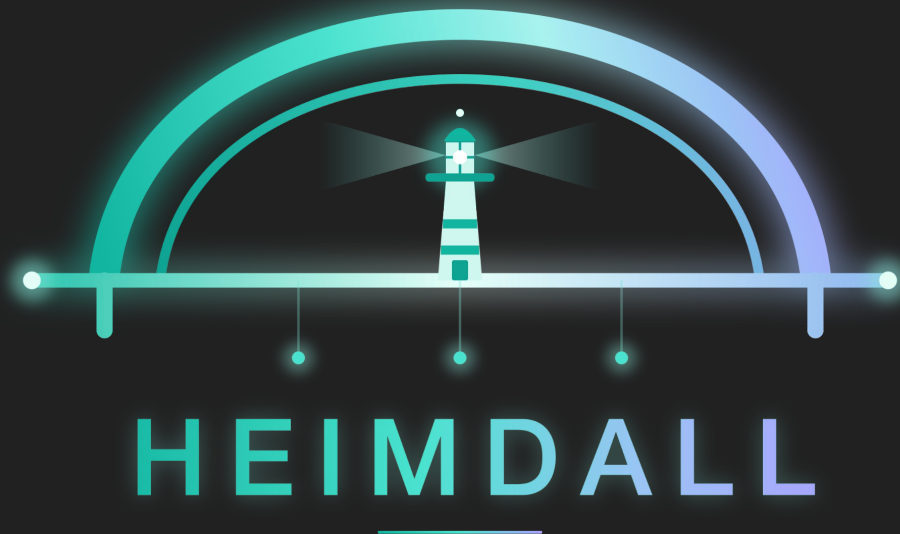


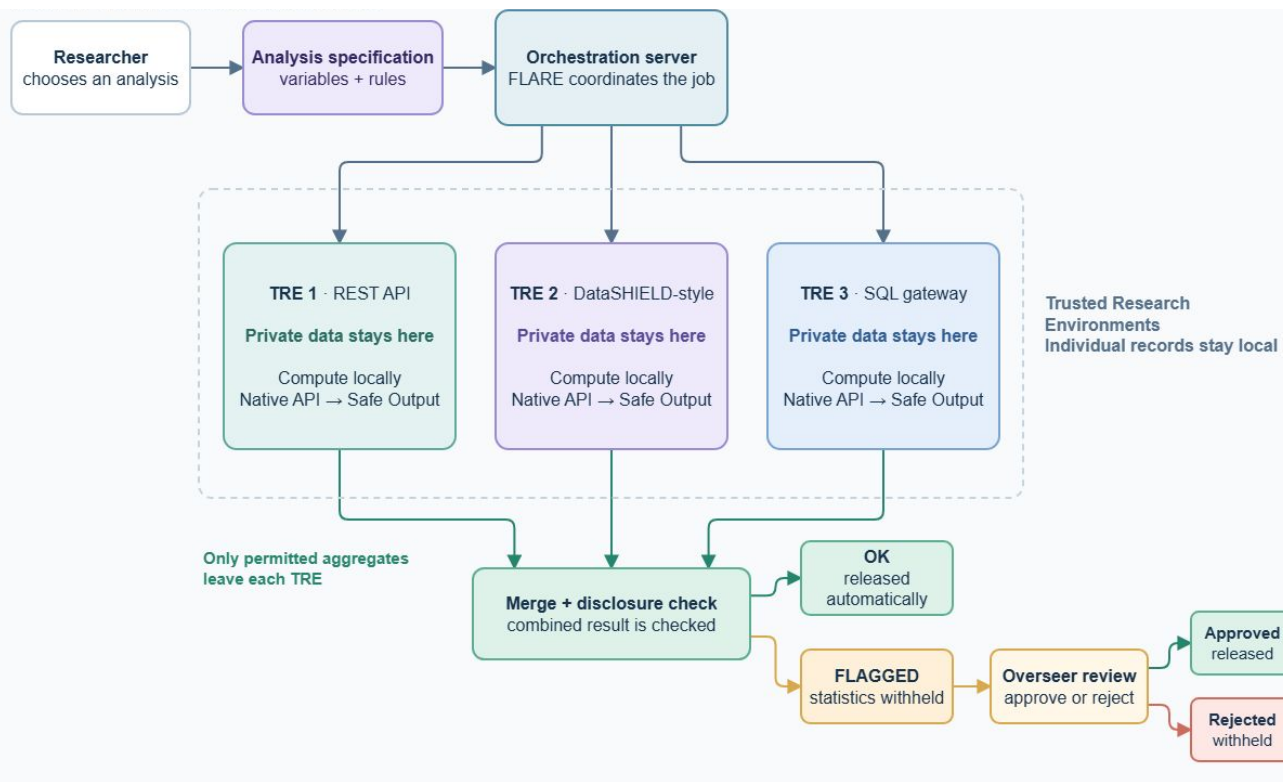


Team:

Espen Hagen, Ioannis Christofilogiannis, Vitalii Babenko, Gaurang Sharma, Melissa Wong, Pedro Gabriel Campana, Marta Menta Czinkoczky, Udogwu Emiri

Federated analysis across TREs with different APIs

Team 1 · Nordic Biobank X NVIDIA 2026 hackathon



Request

type, canonical variables, filters, min_cell_size, project_id.



Dispatch

The FLARE controller broadcasts the spec and collects results



Local execution

Executor loads the adapter for its own tre_id, calls the native API.



Safe output

The TRE filters results before passing them to the FLARE client



Aggregation and review

The server combines results and checks disclosure risk.

Different APIs and schemas across three TREs

The analysis must combine results accurately while keeping individual records inside each TRE.



TRE hunt

REST

```
POST /query
{"cols": ["rs001_gt"],
 "filters": {"alder": {">=": 60}},
 "agg": "value_counts"}
```

local columns

alder · kjon · bmi_baseline · rs001_gt



TRE gefion

DataSHIELD-style

```
POST /ds
{"fn": "ds.table",
 "args": {"x": "D$SNP_1"}}

(nfilter.tab = 3 applied
first)
```

local columns

age_years · sex_male · body_mass_index · SNP_1



TRE brev

SQL gateway

```
POST /sql
{"sql": "SELECT g_rs001,
count(*)
      FROM cohort
      GROUP BY
g_rs001"}
```

local columns

AGE · SEX · BMI · g_rs001

Research questions: allele frequency of rs001 across all cohorts, and SBP ~ age + sex + BMI + LDL + 3 SNPs. Both use canonical variable names.

TRE-Adapter

Each adapter translates the local API response into an `AggregateResult`.

```
class TREAdapter(ABC):
    def schema(self)                # canonical ->
dtype
    def count(self, filters) -> int
    def describe(self, col, filters) # n sum sum_sq min
max
    def value_counts(self, col, filters) # {level: n}
    def gram(self, cols, filters)      # [1,y,X]^T
[1,y,X]

    def run(self, spec) -> AggregateResult
    # shared, never overridden:
    #   canonical -> local (canonical.yaml)
    #   primitives in the TRE's dialect
    #   Safe Output filter + audit line

registry.load("gefion") # sites.yaml -> datashield
```



API implementations

rest.py, datashield.py and sql.py are 40–60 lines each. DataSHIELD nfilter.tab suppressions retain the label `tre_native`.



Regression uses `gram()`

Federated linear regression (M4) reuses `gram()`, which also supports means and variances. The existing adapters needed no changes.



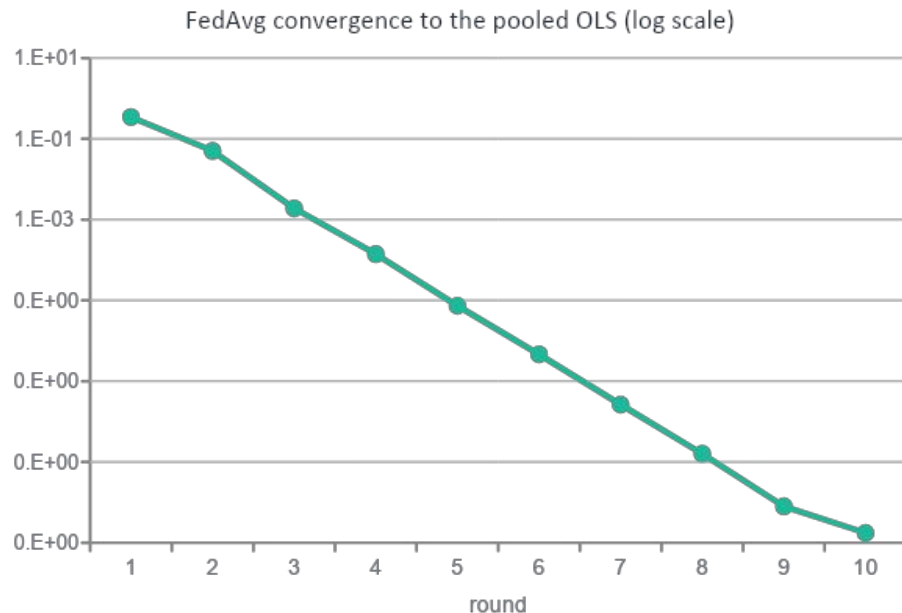
Runtime support

The adapter, executor and job builder run in unit tests, the FLARE simulator, separate FLARE processes and Docker containers.

Use case - Linear regression

$$y \sim \text{age} + \text{sex} + \text{bmi} + \text{ldl} + \text{rs001} + \text{rs007} + \text{rs013}$$

coefficient	exact (1 round)	FedAvg (10 rounds)	pooled OLS
intercept	89.2729	89.2729	89.2729
age	0.5055	0.5055	0.5055
sex	4.2034	4.2034	4.2034
bmi	0.8062	0.8062	0.8062
ldl	1.2189	1.2189	1.2189
snp_rs001	2.3934	2.3934	2.3934
snp_rs007	-1.2817	-1.2817	-1.2817
snp_rs013	0.9003	0.9003	0.9003
max error	1.7e-11	1.8e-11	—



Testing and deployment status: 3 APIs

Cloud demo, TRE demo on-device, UI proof of concept



In-process

Unit tests:
adapters, filter,
merge, disclosure
check, overseer



FLARE simulator

3 to 100 simulated
TREs, with exact
results at each
tested size



Real FLARE processes

Server and 3 clients
using mTLS gRPC and
admin-API submission.
With one client
stopped, 2 sites
reported.



Docker

7 images on isolated
networks.
Jobs run inside
flare-server.
With one container
stopped, 2 sites
reported.

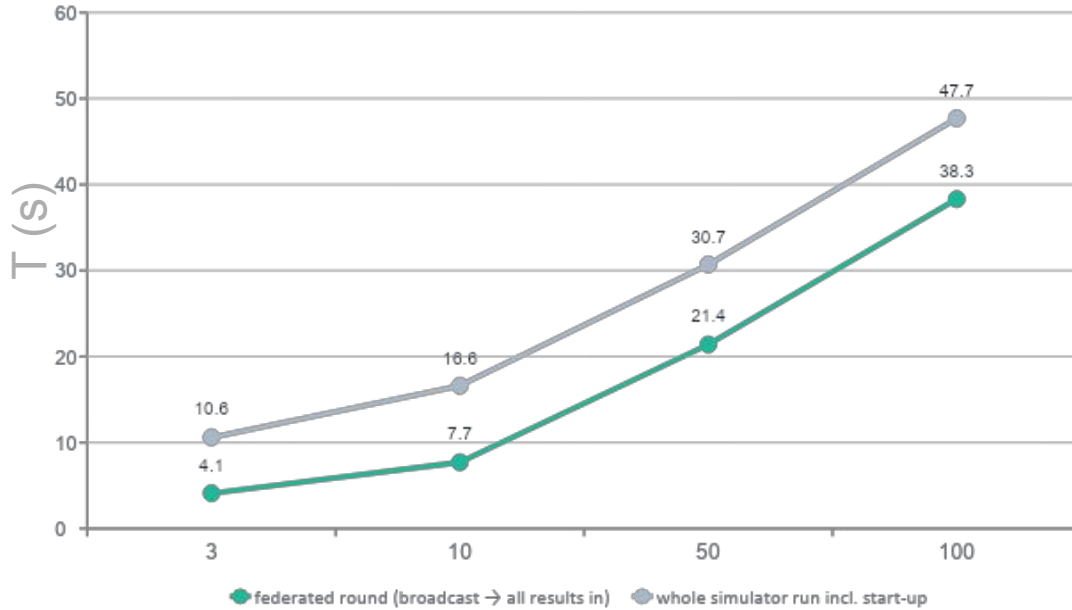


next

Cloud

Clients on HUNT
Cloud, Gefion and
NextCloud, server on
Brev/AWS.
Each TRE needs an
outbound firewall
rule.

Simulated Scaling test: 3 to 100 TREs



100/100

sites reported, with an error of 0.0

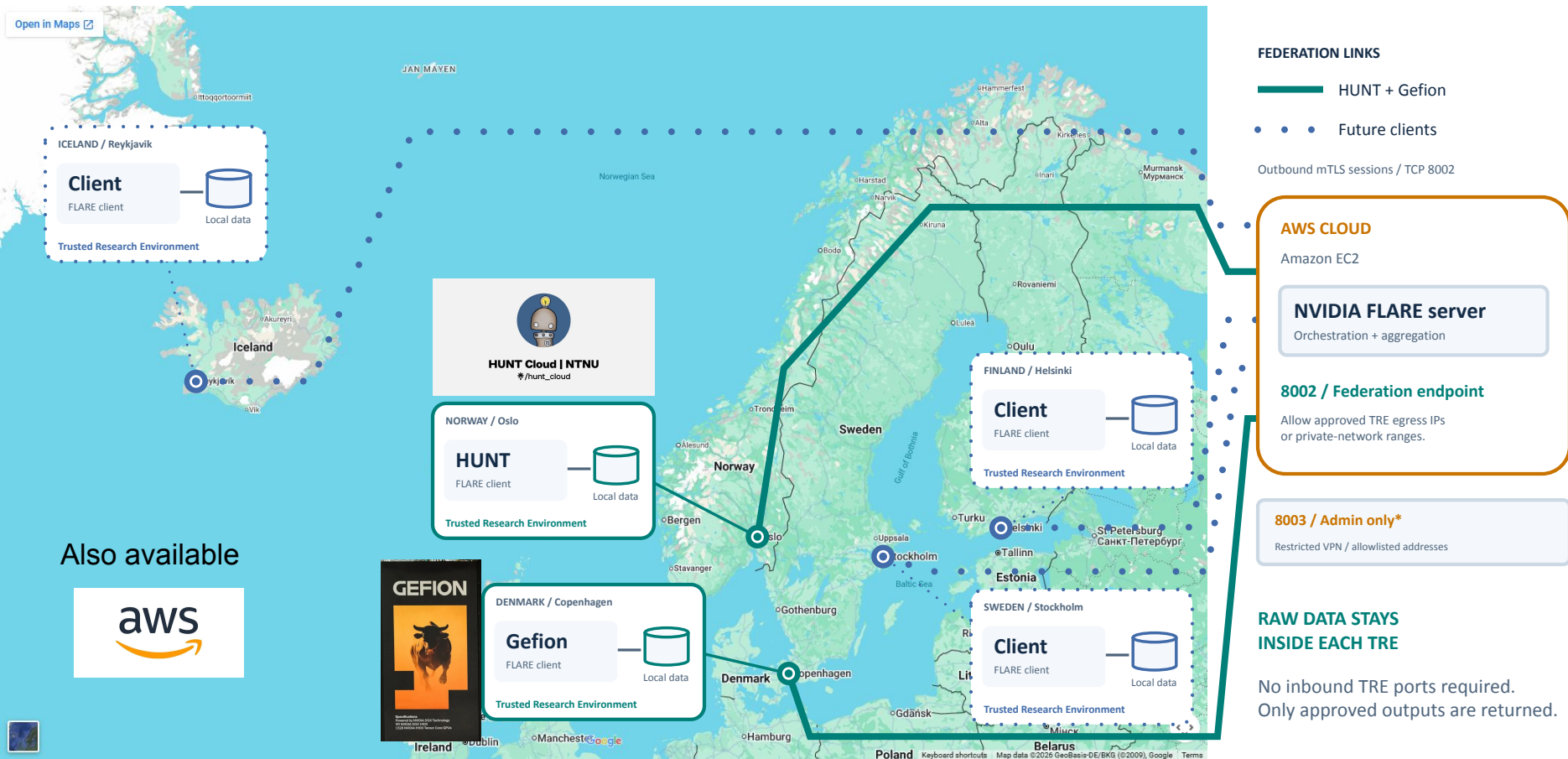
< 1 ms

to sum results from 100 sites

1.4 s

to add a TRE and generate its configuration and kit

Nordic TREs, server and admin on AWS




```

CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS      PORTS          NAMES
ubuntu@ip-172-31-26-165:~/workspace/example_project/prod_00/server1$ docker run --rm -it --name nvflare-server --user "$(id -u):$(id -g)" -p 8002:8002 -p 8002:8002 -v "$PWD":/workspace -v "$PWD/runtime":/tmp/nvflare -w /workspace nvflare:2.9.0 bash startup/sub_start.sh
WORKSPACE set to /workspace
PYTHONPATH is /local/custom:
start fl because process of 16 does not exist
new pid 17
2026-09-17 15:20:02,142 - ServerDeployer - INFO - server heartbeat timeout set to 600
2026-09-17 15:20:02,240 - CoreCell - INFO - server: creating listener on http://0:8002?admin_listener=true
2026-09-17 15:20:02,580 - CoreCell - INFO - server: created backbone external listener for http://0:8002?admin_listener=true
2026-09-17 15:20:02,581 - conn_manager - INFO - Connector [CH00002 PASSIVE tcp://127.0.0.1:36551] is starting
2026-09-17 15:20:03,083 - CoreCell - INFO - server: created backbone internal listener for tcp://localhost:36551
2026-09-17 15:20:03,083 - conn_manager - INFO - Connector [CH00001 PASSIVE http://0:8002?admin_listener=true] is starting
2026-09-17 15:20:03,085 - Cell - INFO - Register blob CB for channel='aux_communication', topic='*'
2026-09-17 15:20:03,086 - FederatedServer - INFO - max_reg_duration=60.0
2026-09-17 15:20:03,087 - AuxRunner - INFO - registered aux handler for topic ObjectStreamer.Request
2026-09-17 15:20:03,088 - AuxRunner - INFO - registered aux handler for topic ObjectStreamer.Abort
2026-09-17 15:20:03,089 - AuxRunner - INFO - registered aux handler for topic fed.event
2026-09-17 15:20:03,091 - ServerDeployer - INFO - deployed FLARE Server.
2026-09-17 15:20:03,097 - Cell - INFO - Register blob CB for channel='hci_channel', topic='*'
2026-09-17 15:20:03,098 - hci - INFO - Admin Server is started
2026-09-17 15:20:03,099 - root - INFO - Server started
2026-09-17 15:26:46,170 - conn_manager - INFO - Connection [CN00002 172.17.0.2:8002 <= 172.31.16.187:38904 SSL site-2] is created: PID: 17
2026-09-17 15:26:46,368 - ClientManager - INFO - Client: New client site-2@172.17.0.2 joined. Sent token. Total clients: 1
2026-09-17 15:26:46,369 - FederatedServer - INFO - client site-2 registered with site_config keys: ['allow_log_streaming']

```

SERVER (AWS)

CLIENT (HUNT Cloud)

```

(team1) ubuntu@ncfh26-hackathon-iaas-aLula:~/users/espen.hagen/site-1/startup$ grep -q "server1" /etc/hosts || echo "16.170.143.132 server1"
; ./sub_start.sh --once
16.170.143.132 server1
WORKSPACE set to /home/ubuntu/users/espen.hagen/site-1
PYTHONPATH is /local/custom:
Start NVFlare directly
2026-09-18 12:41:19,575 - AuxRunner - INFO - registered aux handler for topic ObjectStreamer.Request
2026-09-18 12:41:19,576 - AuxRunner - INFO - registered aux handler for topic ObjectStreamer.Abort
2026-09-18 12:41:19,745 - CoreCell - INFO - site-1: created backbone external connector to http://server1:8002
2026-09-18 12:41:19,746 - conn_manager - INFO - Connector [CH00002 PASSIVE tcp://127.0.0.1:58822] is starting
2026-09-18 12:41:20,247 - CoreCell - INFO - site-1: created backbone internal listener for tcp://localhost:58822
2026-09-18 12:41:20,248 - conn_manager - INFO - Connector [CH00001 ACTIVE http://server1:8002] is starting
2026-09-18 12:41:20,250 - Cell - INFO - Register blob CB for channel='server_command', topic='get_task'
2026-09-18 12:41:20,251 - Cell - INFO - Register blob CB for channel='server_command', topic='submit_update'
2026-09-18 12:41:20,251 - FederatedClient - INFO - Wait for engine to be created.
2026-09-18 12:41:20,333 - conn_manager - INFO - Connection [CN00002 10.5.5.115:46984 => 16.170.143.132:8002 SSL server1]
2026-09-18 12:41:20,576 - Cell - INFO - Register blob CB for channel='aux_communication', topic='*'
2026-09-18 12:41:20,578 - AuxRunner - INFO - registered aux handler for topic fed.event
2026-09-18 12:41:20,685 - Authenticator - INFO - verified server identity 'server1'
2026-09-18 12:41:20,804 - Authenticator - INFO - Verified received token and signature successfully
2026-09-18 12:41:20,805 - FederatedClient - INFO - Successfully registered client:site-1 for project example_project.
2026-09-18 12:41:20,807 - FederatedClient - INFO - Got engine after 0.5561730861663818 seconds
2026-09-18 12:41:20,808 - FederatedClient - INFO - Connected to server: http://server1:8002

```

Swagger - User endpoint for aggregated results

127.0.0.1:8500/docs#/

Bifrost federated analysis API

0.2.0

OAS 3.1

[/openapi.json](#)

Bifrost HTTP API -- the contract the researcher UI (frontend) talks to.

`SERVER_OUT=server/api_out uvicorn server.api:app --port 8500 --workers 1`

Start mock TREs separately with `python scripts/dev_tres.py`. Neither TREs nor FLARE are required for startup or /health. API_TRE_TIMEOUT (seconds per HTTP operation, default 10) and API_TRE_WORKERS (default 8) control adapter fan-out. API_CORS_ORIGINS (comma-separated, default http://localhost:3000) allows the UI origin. Use an exclusive SERVER_OUT directory; see analysis_service's concurrency and demo-versus-secure-deployment notes. Variable names are canonical (snp_rs001), never local column names; requests carry variable names, never participant data. Response shapes: server/schemas.py, exported to docs/schemas/ and served at /openapi.json.

default

GET

/health

Health

GET

/allele-frequency

Allele Frequency

POST

/linear-regression

Linear Regression

POST

/logistic-regression

Logistic Regression

GET

/metadata

Metadata

GET

/examples/{name}

Example

POST

/run

Submit Run

GET

/run/{run_id}

Run Status

GET

/overseer

Overseer

POST

/overseer/{spec_hash}/{decision}

Overseer Decide

GET

/audit

Audit

Schemas

AlleleCounts

>

Expand all

object

Use case 1 - calculate allele freq

GET /allele-frequency Allele Frequency

Parameters

Name	Description
variant * required	
string (query)	snp_rs001
	minLength: 1
project_id * required	
string (query)	ncfh-2026-demo
	minLength: 1
min_cell_size	
integer (query)	5
	minimum: 5

Input

- Variant
- Project id

```
{
  "rejected_per_site": {},
  "stats": {
    "snp_rs001": {
      "genotype_counts": {
        "0": 25690,
        "1": 4143,
        "2": 157
      },
      "allele_counts": {
        "minor": 4477,
        "major": 55523,
        "n_alleles": 60000
      },
      "allele_freq": 0.07461666666666666,
      "allele_freq_sites": [
        "brev",
        "gefion",
        "hunt"
      ]
    }
  }
}
```

Output (json)

- Allele frequency

Use case 2 - linear regression (SBP ~ age + BMI)

POST /linear-regression Linear Regression

Parameters

No parameters

Request body * required

Edit Value | Schema

```
{
  "features": ["age", "bmi"],
  "target": "sbp",
  "project_id": "ncfh-2026-demo",
  "filters": {},
  "min_cell_size": 5
}
```

Input (json)

- Outcome + predictor variables

Output (json)

- Regression coefficients

```
{
  "region_per_site": {
    "hunt": "nordic",
    "gefion": "nordic",
    "brev": "nordic"
  },
  "rejected_per_site": {},
  "stats": {
    "_linreg": {
      "ols": {
        "outcome": "sbp",
        "coef": {
          "intercept": 95.52337125694957,
          "age": 0.5058367368789758,
          "bmi": 0.8092701406242438
        }
      },
      "n": 30000
    }
  }
}
```

Preliminary User Experience: What we suggest for the future

The screenshot displays the BIFROST Federated analysis gateway interface. The browser address bar shows `http://localhost:3000`. The header bar is dark blue with the BIFROST logo and text "BIFROST Federated analysis gateway". On the right of the header, there is a "Secure session" button and a user profile icon labeled "RW".

On the left, a "RESEARCHER MENU" is visible with the following items:

- Analysis (highlighted)
- Overseer queue
- Audit logs

The main content area is titled "BIFROST / ANALYSIS WORKSPACE" and features a large heading "Cross-TRE analysis". Below this heading is a subtext: "Build a request, run it across connected environments, and inspect disclosure-safe output." A button labeled "Copy result JSON" is located to the right of the subtext.

A yellow informational banner states: "Record-level data stays inside each TRE. Only aggregate statistics and model information pass through the gateway."

Below the banner, two analysis options are presented:

- Allele frequency** (Federated analysis) - This option is highlighted with a blue border.
- Linear regression** (Federated learning)

The interface is divided into two main sections for the selected analysis:

STEP 1: Define your analysis

This section includes a "Load example..." dropdown menu. Below it, a "Safe project" dropdown menu is set to "ncfh-2026-demo". Under "Genotype variables", there are checkboxes for "snp_rs001" (checked), "snp_rs002", and "snp_rs003".

STEP 2: Connected TREs

This section shows a status of "3 online". It lists two connected TREs:

- hunt** (rest - NO) with a status of "Ready".
- gefion** (datashield - NO) with a status of "Ready".

A red circular notification icon with the letter "N" is located in the bottom left corner of the interface.

```

CONTAINER ID   IMAGE      COMMAND                  STATUS    PORTS       NAMES
ubuntu@ip-172-31-26-165:~/workspace/example_project/prod_00/server1$ docker run --rm -it --name nvflare-server --user "$(id -u):$(id -g)" -p 8002:8002 -p 8003:8003 -v "$PWD":/workspace -v "$PWD/runtime":/tmp/nvflare -w /workspace nvflare:2.9.0 bash startup/sub_start.sh
WORKSPACE set to /workspace
PYTHONPATH is /local/custom:
start fl because process of 16 does not exist
new pid 17
2026-09-17 15:20:02,142 - ServerDeployer - INFO - server heartbeat timeout set to 600
2026-09-17 15:20:02,240 - CoreCell - INFO - server: creating listener on http://0:8002?admin_listener=true
2026-09-17 15:20:02,580 - CoreCell - INFO - server: created backbone external listener for http://0:8002?admin_listener=true
2026-09-17 15:20:02,581 - conn_manager - INFO - Connector [CH00002 PASSIVE tcp://127.0.0.1:36551] is starting
2026-09-17 15:20:03,083 - CoreCell - INFO - server: created backbone internal listener for tcp://localhost:36551
2026-09-17 15:20:03,083 - conn_manager - INFO - Connector [CH00001 PASSIVE http://0:8002?admin_listener=true] is starting
2026-09-17 15:20:03,085 - Cell - INFO - Register blob CB for channel='aux_communication', topic='*'
2026-09-17 15:20:03,086 - FederatedServer - INFO - max_reg_duration=60.0
2026-09-17 15:20:03,087 - AuxRunner - INFO - registered aux handler for topic ObjectStreamer.Request
2026-09-17 15:20:03,088 - AuxRunner - INFO - registered aux handler for topic ObjectStreamer.Abort
2026-09-17 15:20:03,089 - AuxRunner - INFO - registered aux handler for topic fed.event
2026-09-17 15:20:03,091 - ServerDeployer - INFO - deployed FLARE Server.
2026-09-17 15:20:03,097 - Cell - INFO - Register blob CB for channel='hci_channel', topic='*'
2026-09-17 15:20:03,098 - hci - INFO - Admin Server is started
2026-09-17 15:20:03,099 - root - INFO - Server started
2026-09-17 15:26:46,170 - conn_manager - INFO - Connection [CN00002 172.17.0.2:8002 <= 172.31.16.187:38904 SSL site-2] is created: PID: 17
2026-09-17 15:26:46,368 - ClientManager - INFO - Client: New client site-2@172.17.0.2 joined. Sent token. Total clients: 1
2026-09-17 15:26:46,369 - FederatedServer - INFO - client site-2 registered with site_config keys: ['allow_log_streaming']

```

SERVER (AWS)

CLIENT (AWS)

```

ubuntu@ip-172-31-16-187:~/nvflare-docker/site-2$ sudo docker run --rm -it --name site-2 \
--add-host server1:172.31.26.165 \
-v "$PWD":/workspace -v "$PWD/runtime":/tmp/nvflare -w /workspace \
nvflare:2.9.0 bash startup/sub_start.sh
WORKSPACE set to /workspace
PYTHONPATH is /local/custom:
start fl because of no pid.fl
new pid 13
2026-09-17 15:26:45,240 - AuxRunner - INFO - registered aux handler for topic ObjectStreamer.Request
2026-09-17 15:26:45,241 - AuxRunner - INFO - registered aux handler for topic ObjectStreamer.Abort
2026-09-17 15:26:45,600 - CoreCell - INFO - site-2: created backbone external connector to http://server1:8002
2026-09-17 15:26:45,602 - conn_manager - INFO - Connector [CH00002 PASSIVE tcp://127.0.0.1:36640] is starting
2026-09-17 15:26:46,103 - CoreCell - INFO - site-2: created backbone internal listener for tcp://localhost:36640
2026-09-17 15:26:46,104 - conn_manager - INFO - Connector [CH00001 ACTIVE http://server1:8002] is starting
2026-09-17 15:26:46,105 - Cell - INFO - Register blob CB for channel='server_command', topic='get_task'
2026-09-17 15:26:46,106 - Cell - INFO - Register blob CB for channel='server_command', topic='submit_update'
2026-09-17 15:26:46,106 - FederatedClient - INFO - Wait for engine to be created.
2026-09-17 15:26:46,170 - conn_manager - INFO - Connection [CN00002 172.17.0.2:38904 => 172.31.26.165:8002 SSL server1] is created: PID: 13
2026-09-17 15:26:46,242 - Cell - INFO - Register blob CB for channel='aux_communication', topic='*'
2026-09-17 15:26:46,243 - AuxRunner - INFO - registered aux handler for topic fed.event
2026-09-17 15:26:46,321 - Authenticator - INFO - verified server identity 'server1'
2026-09-17 15:26:46,372 - Authenticator - INFO - Verified received token and signature successfully
2026-09-17 15:26:46,373 - FederatedClient - INFO - Successfully registered client:site-2 for project example_project.
2026-09-17 15:26:46,378 - FederatedClient - INFO - Got engine after 0.2717094421386719 seconds
2026-09-17 15:26:46,379 - FederatedClient - INFO - Connected to server: http://server1:8002

```

Next steps



Cloud deployment

Configure the server-client setup with real TREs



Researcher interface

Build a page for selecting variables and running analyses, with results, coverage and audit records.



Real variable names and variants

Replace the invented local column names and rs001–rs020 with each site's schema and real rsIDs



Hierarchical aggregation

Add regional relay aggregators through FLARE's hierarchical topology, using the existing associative merge.

Thank You!

